SUPPLEMENTARY MATERIAL: WHEN DOES CSP HELP A WELL-TRAINED SURROGATE?

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ABSTRACT

This document accompanies the ICASSP 2027 paper "Physics-informed but not physics-consistent" and reports results obtained after submission. The paper evaluates calibrated solution-subspace projection (CSP) on gridfm-graphkit checkpoints trained for 120 epochs. Here we ask what happens when the same surrogate is trained further, and when CSP should be applied at all. Three findings qualify the paper's results. (i) After CSP, Mean PB lands close to a model-independent *projection floor*, the PB of the projected Newton reference. Whether the raw Mean PB lies above that floor predicts whether CSP_k lowers it on every cross-grid case with $16 \le k \le 32$ and $k < N$. (ii) Continued training drives raw PB toward and below the floor, so the benefit of CSP_{16} shrinks and can reverse; on GBnetwork it turns from a reduction into an increase. (iii) On large grids, CSP trades voltage-magnitude accuracy for power-balance consistency. The projection rank controls this trade-off. Projecting the angle alone does not keep the PB gain, and magnitude-only projection helps only when the angle is already more accurate than its projected reference.

S1. SETUP AND DEFINITIONS

Protocol. All numbers use the scorer of the paper's Table 1: seed-42 1:1:1 split, train-only calibration and SVD bases, full-state projection without restoring known setpoints, and complex128 Mean/Max PB on a 100-MVA base. On the paper's seed-42 gridfm-graphkit checkpoint (identical SHA-256), the two scorer versions agree to $\sim 10^{-7}$ relative in all mean metrics. Unlike Table 1, every result here uses a single model seed (42).

Models. gridfm-graphkit with the released configuration (12 layers, hidden 48, 8 heads) and the stable initialization of the paper, trained with LR 10^{-4} for 120 epochs. "Converged" models continue from that checkpoint for 200 further epochs at a lower learning rate.

Projection floor. For rank k, let $x^\star_{P_k} = \bar{x}_{\text{tr}} + P_k(x^\star - \bar{x}_{\text{tr}})$ be the projected Newton reference. Its Mean PB,

$$\text{PB}_k^{\text{floor}} = \text{MeanPB}\big(x^\star_{P_k}\big), \tag{S1}$$

does not depend on the surrogate. It is the PB that projection leaves even for a perfect prediction. When the calibrated error is mostly off-subspace, CSP_k maps the prediction close to $x^\star_{P_k}$, so its Mean PB lands near the floor. The prediction is therefore improved only if its raw PB lies above the floor (cf. Eq. (4) of the paper for the corresponding statement on state accuracy).

Projection channels. CSP_k projects magnitude and angle together. $\text{CSP}_{V,k}$ projects only the calibrated magnitude and keeps the calibrated angle; $\text{CSP}_{\theta,k}$ projects only the angle and keeps the calibrated magnitude. The floors $\text{PB}_k^{\text{floor,mag}}$ and $\text{PB}_k^{\text{floor,ang}}$ apply the same partial projection to the reference.

Table S1. Training longer moves CSP_{16} toward a loss on every grid. Left: metrics of the raw prediction. Right: Mean PB of C and of CSP_{16} divided by raw Mean PB (< 1: lower PB than raw), and raw Mean PB divided by the rank-16 floor (> 1: CSP can lower PB). *LVN_heo1: Kron-reduced model (328 pass-through buses eliminated exactly), scored on all 722 buses; Sec. S4 uses the unreduced model.

		Raw prediction			Mean PB ratio		
Grid	Ep.	V RMSE	a_{w}	Mean PB	C/raw	CSP_{16}/raw	raw/floor
case89pegase	120	.00176	.991	.0476	0.74	5.99	0.17
	+200	.00056	.996	.0187	0.88	15.21	0.07
case118	120	.00037	.999	.0137	0.87	7.22	0.14
	+200	.00019	1.000	.0067	0.96	14.71	0.07
case300	120	.00643	.969	.1102	0.92	1.16	0.89
	+200	.00433	.981	.0744	0.97	1.68	0.60
LVN_heo1*	120	.00221	.997	.0092	0.81	0.53	2.70
	+200	.00179	.990	.0045	0.90	0.87	1.33
case1354pegase	120	.00483	.932	.1317	0.92	1.37	0.73
	+200	.00300	.975	.0966	0.95	1.87	0.53
case1888rte	120	.00691	.870	.1418	0.89	0.38	2.70
	+200	.00587	.905	.0870	0.93	0.60	1.66

S2. CONVERGED GRIDFM-GRAPHKIT ON SIX GRIDS

Starting from the 120-epoch checkpoints, we trained for 200 further epochs at LR 3×10^{-5} on six grids. At 120 epochs, these grids span both outcomes of CSP_{16}. On case89pegase, case118, case300 and case1354pegase, CSP_{16} raised Mean PB; on LVN_heo1 and case1888rte, it lowered it.

Raw accuracy improves; the CSP benefit shrinks or reverses. Longer training lowers raw V RMSE and raw Mean PB on all six grids, while the rank-16 floor does not depend on the model. Raw PB over the floor therefore falls on every grid (Table S1). Where CSP_{16} already raised PB at 120 epochs, the increase grows: on case118 from $7.2\times$ to $14.7\times$ raw, and on case89pegase from $6.0\times$ to $15.2\times$. Where it helped, the benefit shrinks: on case1888rte from $0.38\times$ to $0.60\times$ raw. LVN_heo1 behaves the same way: training halves its raw PB ($.0092 \to .0045$), raw/floor falls from 2.70 to 1.33, and the CSP_{16} ratio rises from $0.53\times$ to $0.87\times$ raw.

Calibration remains safe but weakens. C lowers Mean PB on every grid at both training lengths, but by less after convergence (e.g., case89pegase $0.74 \to 0.88$ of raw). Longer training itself removes much of the systematic per-bus bias. C leaves R^2_{w} and a_{w} unchanged by construction. The same holds across all 30 cross-grid systems (Sec. S4), where C lowered Mean PB, V RMSE and θ RMSE on

Table S2. Projection channels for the converged models. All entries are Mean-PB ratios to raw except the last (V RMSE of CSP_k over raw). F, F_V, F_θ: floors for full, magnitude-only and angle-only projection of the reference. The high rank is 64 when $N \leq 128$. *LVN_heo1: Kron-reduced model (328 pass-through buses eliminated exactly), scored on all 722 buses; Sec. S4 uses the unreduced model.

Grid	k	C	CSP	CSP_V	CSP_θ	F	F_V	F_θ	V
case89pegase	16	0.88	15.21	2.89	14.24	15.21	2.68	14.11	2.30
	64	0.88	1.11	1.02	1.01	0.25	0.14	0.16	0.88
case118	16	0.96	14.71	19.90	20.76	14.70	19.84	20.70	94.79
	64	0.96	4.07	1.73	3.58	3.87	1.26	3.32	2.04
case300	16	0.97	1.68	1.80	2.00	1.67	1.45	1.81	3.59
	128	0.97	1.21	1.01	1.22	0.85	0.31	0.71	0.94
LVN_heo1*	16	0.90	0.87	0.85	1.26	0.75	0.29	0.69	0.77
	128	0.90	0.51	0.77	0.86	0.25	0.13	0.18	0.91
case1354pegase	16	0.95	1.87	5.78	6.10	1.87	5.37	5.85	6.03
	128	0.95	3.51	4.76	4.96	3.21	4.26	4.50	3.30
case1888rte	16	0.93	0.60	0.97	1.06	0.60	0.47	0.70	2.17
	128	0.93	0.74	0.87	1.01	0.61	0.30	0.54	1.63

every grid but left Max PB unchanged (median ratio 1.00).

No projection channel recovers the loss. Table S2 compares full, magnitude-only (CSP_V) and angle-only (CSP_θ) projection. On the four grids where raw PB is below the floor, every channel raises PB at $k = 16$. At the higher rank, no channel beats C on any of the four. The loss shrinks on case89pegase, case118 and case300, but full projection on case1354pegase loses more as the rank grows. On the two grids above the floor, full projection is the best channel, except on LVN_heo1 at $k = 16$, where magnitude-only projection is marginally better (0.85 vs. 0.87 of raw). These results are consistent with the 120-epoch cross-grid study (Sec. S4). The PB gain of CSP requires projecting both channels jointly, and it exists only while the raw prediction is above the corresponding floor.

Rank does not rescue the grids below the floor. Table S3 extends Table S2 to every rank. Relative to C, a projection improves Mean PB only on case1888rte and LVN_heo1, the two grids whose raw PB is still above the floor. On the other four grids every channel is worse than C at every rank with $k < N$. On case89pegase, case118 and case300 the loss shrinks as k approaches N, where projection becomes the identity. On case1354pegase it grows with k. The two improving grids also behave differently with rank. On case1888rte the gain of full CSP is largest at $k = 16$ (-36.0%) and decreases to -21.0% at $k = 128$. On LVN_heo1 it increases with rank (-2.5% to -43.6%). Here convergence removes mainly the low-rank gain: at 120 epochs the same model improved on C by 34.1% at $k = 16$ and 44.7% at $k = 128$. A plausible reading is that longer training fits the leading solution directions, so only the finer directions kept at high rank still carry correctable error. Rank must therefore be chosen per grid on validation data; no single rank works for every converged model.

S3. GBNETWORK: TRAINING LONGER

The paper's Table 1 uses three 120-epoch gridfm-graphkit seeds. Here we continue the seed-42 checkpoint (identical SHA-256) for 200 epochs at three learning rates and rescore with the same protocol (Table S4). For seed 42, CSP_{16} lowers Mean PB by 34.9% relative to C; the paper reports 40.5% for the three-seed mean.

The CSP_{16} gain disappears once raw PB passes the floor. At LR 3×10^{-5}, training lowers raw Mean PB from .0792 to .0380, below the rank-16 floor of .0415. CSP_{16} then increases Mean PB by 22.4% relative to C (Table S5). The other two runs stop at raw PB .0565 (LR 10^{-4}, still above the floor; CSP_{16} -15.9%) and .0411 (LR 10^{-5}, at the floor; CSP_{16} $+8.9\%$). The sign of the change follows raw PB relative to the floor, as predicted by Eq. (S1).

Table S3. Converged models (+200 ep) at every rank: Mean PB of raw and C (10^{-3} p.u.), change (%) of CSP_k, $\text{CSP}_{V,k}$ and $\text{CSP}_{\theta,k}$ relative to C (bold: best channel when it improves on C), and the full and magnitude-only floors (10^{-3}). Same layout as Table S5. *Kron-reduced LVN_heo1.

Grid	k	Raw	C	CSP	CSP_V	CSP_θ	F	F_V
case89pegase	16	18.7	16.4	+1627.9	+228.2	+1517.3	284	50.1
	32	18.7	16.4	+308.6	+80.1	+264.2	64.4	21.9
	64	18.7	16.4	+26.3	+15.4	+14.6	4.62	2.57
	128			$k \geq N$: P_k is the identity, CSP$\equiv C$				
case118	16	6.68	6.44	+1427.0	+1965.6	+2054.7	98.3	133
	32	6.68	6.44	+1419.1	+1208.2	+1227.3	97.6	83.7
	64	6.68	6.44	+322.3	+79.9	+271.4	25.9	8.43
	128			$k \geq N$: P_k is the identity, CSP$\equiv C$				
case300	16	74.4	72	+73.0	+86.4	+106.7	124	108
	32	74.4	72	+78.7	+66.9	+86.4	125	91.1
	64	74.4	72	+72.7	+28.0	+61.6	116	55.5
	128	74.4	72	+24.9	+4.2	+26.0	63	23.4
LVN_heo1*	16	4.52	4.05	-2.5	**-5.4**	+40.6	3.39	1.29
	32	4.52	4.05	**-17.0**	-9.7	+24.5	2.7	1.09
	64	4.52	4.05	**-31.1**	-16.7	+13.6	2.25	0.887
	128	4.52	4.05	**-43.6**	-14.3	-4.0	1.15	0.6
case1354pegase	16	96.6	91.4	+98.1	+510.6	+544.2	181	518
	32	96.6	91.4	+127.6	+502.0	+542.8	201	509
	64	96.6	91.4	+198.3	+463.7	+475.8	256	472
	128	96.6	91.4	+270.7	+402.7	+423.6	310	411
case1888rte	16	87	81.2	**-36.0**	+3.7	+13.8	52.5	41.3
	32	87	81.2	**-30.3**	+0.4	+14.2	54.7	37.2
	64	87	81.2	**-23.9**	-3.5	+13.5	53.8	30.2
	128	87	81.2	**-21.0**	-6.8	+7.9	52.6	26.4

Magnitude-only projection is the remaining useful channel for the best-trained model. The converged angle error (.224° after C) is already below the angle error that rank-16 projection leaves in the reference itself (.386°). Projecting the angle therefore discards valid variation, as Eq. (4) of the paper predicts. The magnitude error (.00327) is still above its projected-reference counterpart (.00258). Consequently, at LR 3×10^{-5} only $\text{CSP}_{V,128}$ improves on C (-5.4%), whereas full and angle-only projection remain worse at every rank. At 120 epochs, by contrast, full projection is best at every rank (-34.9% to -40.4%), and angle-only projection never improves on C for any checkpoint.

S4. GRIDFM-GRAPHKIT ACROSS 31 GRIDS

We retrained gridfm-graphkit with the same recipe on the 31 systems of the paper's cross-grid study and scored it like GridSFM in Fig. 3 of the paper.[1] The GBnetwork entry here comes from this uniform campaign, not from the paper's Table 1 checkpoint.

CSP is not uniformly beneficial for this model. Unlike GridSFM, gridfm-graphkit improves on 19 of 30 grids (Fig. S1, Table S6). It is hurt on several small and medium transmission systems that it already fits closely (e.g., case24, case39, case89pegase, case118, case145), where raw PB lies far below the rank-16 floor.

The floor predicts the sign of the PB change at low rank. Table S7 counts, per rank, the grids with $k < N$, where projection is not the identity. At $k = 16$ and 32, "raw PB above the floor" coincides with "CSP lowers PB" on every such grid. The reason is visible in the

[1] case9241pegase is still training and is omitted. Its first 39 epochs used LR 10^{-4}; the remaining 81 use 3×10^{-5} after an LR-10^{-4} restart diverged. The four largest RTE/ENTSO-E grids were trained as two consecutive 60-epoch runs.

Table S4. GBnetwork gridfm-graphkit, seed 42, $k = 16$; same columns as Table 1 of the paper (inference time omitted).

Checkpoint	Output	V RMSE	θ (°)	R^2_w	a_w	Mean PB	Max PB
120 ep (paper, seed 42)	Raw	.00752	.372	.930	.842	.07917	17.38
	C	.00655	.358	.930	.842	.06841	17.10
	P_{16}	.00544	.495	.965	.832	.04446	4.94
	CSP_{16}	.00539	.490	.965	.832	.04453	5.04
+200 ep, LR 10^{-4}	Raw	.00459	.288	.968	.938	.05649	18.73
	C	.00422	.273	.968	.938	.05020	18.68
	P_{16}	.00389	.457	.975	.927	.04234	4.99
	CSP_{16}	.00386	.451	.975	.927	.04224	5.02
+200 ep, LR 3×10^{-5}	Raw	.00342	.227	.981	.969	.03799	6.10
	C	.00327	.224	.981	.969	.03417	6.01
	P_{16}	.00347	.429	.979	.958	.04184	5.20
	CSP_{16}	.00344	.429	.979	.958	.04183	5.18
+200 ep, LR 10^{-5}	Raw	.00384	.244	.976	.956	.04111	6.35
	C	.00367	.239	.976	.956	.03851	6.33
	P_{16}	.00360	.437	.978	.945	.04194	5.03
	CSP_{16}	.00359	.435	.978	.945	.04194	5.03

Table S5. GBnetwork: Mean PB of raw and C, and the change (%) of CSP_k, magnitude-only $\text{CSP}_{V,k}$ and angle-only $\text{CSP}_{\theta,k}$ relative to C (bold: best channel when it improves on C). F, F_V: floors.

Checkpoint	k	Raw	C	CSP	CSP_V	CSP_θ	F	F_V
120 ep (paper, seed 42)	16	.0792	.0684	**-34.9**	-16.6	+11.3	.0415	.0181
	32	.0792	.0684	**-35.3**	-16.7	+9.0	.0400	.0171
	64	.0792	.0684	**-36.5**	-17.8	+6.5	.0379	.0153
	128	.0792	.0684	**-40.4**	-19.9	+1.4	.0341	.0125
+200 ep, LR 10^{-4}	16	.0565	.0502	**-15.9**	-8.1	+28.2	.0415	.0181
	32	.0565	.0502	**-17.4**	-9.1	+24.7	.0400	.0171
	64	.0565	.0502	**-19.9**	-11.1	+20.5	.0379	.0153
	128	.0565	.0502	**-25.9**	-14.4	+13.2	.0341	.0125
+200 ep, LR 3×10^{-5}	16	.0380	.0342	+22.4	+5.6	+62.7	.0415	.0181
	32	.0380	.0342	+19.2	+3.7	+57.1	.0400	.0171
	64	.0380	.0342	+14.6	+0.2	+50.5	.0379	.0153
	128	.0380	.0342	+5.2	**-5.4**	+39.3	.0341	.0125
+200 ep, LR 10^{-5}	16	.0411	.0385	+8.9	**-0.4**	+50.8	.0415	.0181
	32	.0411	.0385	+6.3	**-1.9**	+46.0	.0400	.0171
	64	.0411	.0385	+2.5	**-4.8**	+40.3	.0379	.0153
	128	.0411	.0385	-5.7	**-9.6**	+30.6	.0341	.0125

decomposition $\hat{x}^{\text{CSP}} = x^\star_{P_k} + P_k e^C$: the CSP output is the projected reference plus the in-subspace part of the calibrated error. At $k = 16$ this part is small, so the output is nearly the projected reference and its Mean PB sits at a median of 1.03× the floor (last column of Table S7). The floor then determines the outcome, and comparing it with raw PB predicts the sign of the change before CSP is applied. As k grows, more of the model's own error passes through the projection, so CSP stays further above the floor (1.38× at $k = 128$), and the rule fails on a few grids. The ratio CSP/floor measures how predictable the effect of CSP is, not how good the result is. The floor itself decreases with k, so CSP can reach a lower Mean PB at high rank while lying further above its floor. On GBnetwork at 120 epochs, for example, CSP_{128} gives .0408 against a floor of .0341 (1.20×), whereas CSP_{16} gives .0445 against .0415 (1.07×).

Rank sets a PB–accuracy trade-off on large grids. Fig. S2 shows the rank dependence for every grid with $N \geq 145$. On the RTE systems (case1888rte, case2848rte, case6470/6495/6515rte), CSP_k lowers Mean PB but raises V RMSE at every rank; on case6470rte the PB ratio goes from 0.20 to 0.31 and the V ratio from 1.60 to 1.23 between $k = 16$ and $k = 128$. On case2869pegase and case3120sp the PB gain disappears by $k = 128$, and case1354pegase is hurt at every rank. Increasing k narrows both the gain and the loss. Two systems behave as in the paper's GBnetwork result and improve both quantities at every rank: GBnetwork and ENTSO-E. On LVN_heo1,

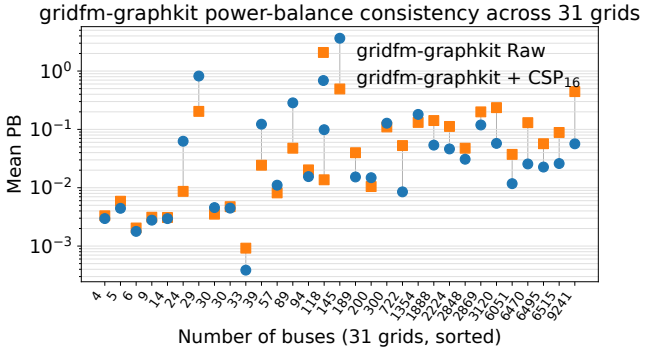


Fig. S1. gridfm-graphkit (120 epochs) Mean PB before and after CSP_{16} on 30 systems ordered by bus count; logarithmic scale. Compare Fig. 3 of the paper (GridSFM).

PB drops at every rank and V RMSE improves slightly.

Angle-only projection does not keep the PB gain. Because the large grids lose magnitude accuracy but gain in angle, we tested $\text{CSP}_{\theta,k}$ on the 12 grids with $N \geq 300$. $\text{CSP}_{\theta,k}$ leaves the magnitude at its calibrated value, so its V RMSE equals that of C. Its Mean PB, however, is higher than that of CSP_k on 12/12 grids at $k = 16$ and higher than raw on 6/12 (green in Fig. S2). The reference shows the same tendency: at $k = 16$, projecting only the angle of $x^\star$ gives a higher PB than projecting both channels on 9/12 grids. The PB gain of CSP therefore comes from projecting both channels jointly, and selecting a single channel cannot avoid the trade-off; the rank remains the control.

S5. PRACTICAL GUIDANCE AND REPRODUCTION

When to apply CSP. (1) Calibration C is safe: it never increased Mean, V, or θ error on any grid or training length studied. Its gain shrinks as training removes the systematic bias, and it does not reduce Max PB. (2) Before deploying CSP_k, compare the validation raw Mean PB with $\text{PB}_k^{\text{floor}}$, which requires only training and validation references. If raw PB is below the floor, projection will increase PB. (3) If PB is above the floor but V RMSE rises, choose k on validation data for the desired balance. Projecting a single channel does not preserve the PB gain.

Reproduction. Result snapshots, checkpoint hashes and job scripts are in `reproduction/results/supplementary_20261002`. All tables are regenerated by `reproduction/code/build_supplementary.py`; the figures by `plot_supplementary_rank_curves.py` and `summarize_gridsfm_31grid_csp16.py` (option `--model-label`).

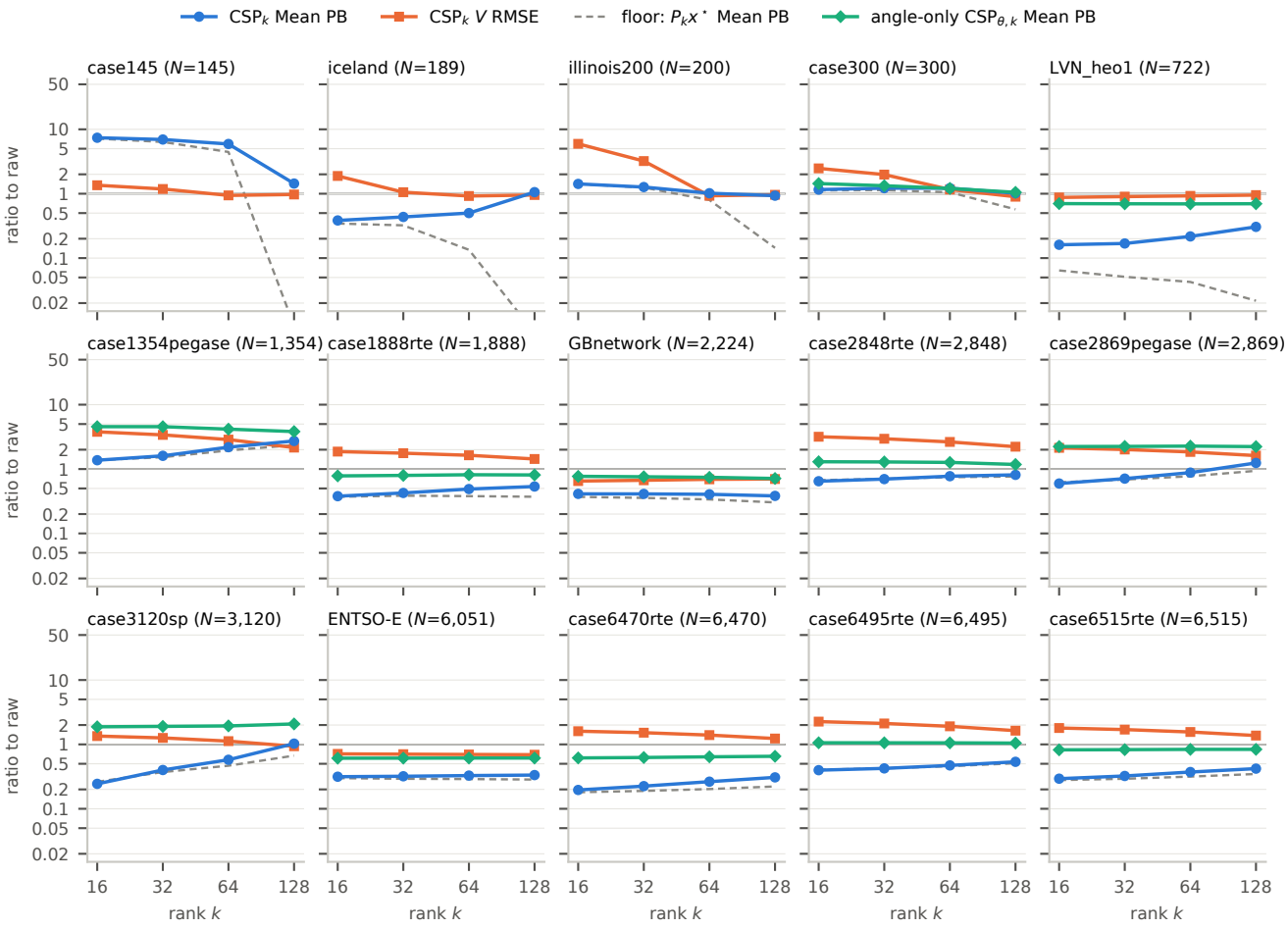


Fig. S2. Rank dependence of CSP for gridfm-graphkit (120 epochs), grids with $N \geq 145$. Ratios to the raw prediction, logarithmic scale; values below 1 improve on raw. Dashed: projection floor $\mathrm{PB}_k^{\mathrm{floor}}$ relative to raw PB.

Table S6. Per-grid gridfm-graphkit at $k = 16$. PB in 10^{-3} p.u.; the last two columns are CSP_{16}/raw ratios. †raw below the floor; $^{\ddagger}N \leq 16$, where P_{16} is the identity and CSP reduces to C.

Grid	N	Raw PB	Floor	CSP PB	PB ratio	V ratio
case4gs‡	4	3.31	–	2.96	0.89	0.96
case5‡	5	5.87	–	4.44	0.76	0.70
case6ww‡	6	2.04	–	1.79	0.87	0.99
case9‡	9	3.12	–	2.78	0.89	0.92
case14‡	14	3.09	–	2.95	0.96	0.87
case24†	24	8.66	61.5	62.7	7.23	1.88
GBreduced†	29	204	808	821	4.03	0.95
case30†	30	3.52	3.72	4.57	1.30	0.90
case_ieee30	30	4.73	2.55	4.47	0.94	0.94
case33bw	33	0.921	0.0685	0.387	0.42	0.94
case39†	39	24.3	121	123	5.05	1.02
case57†	57	8.16	10.3	11.1	1.36	0.86
case89pegase†	89	47.6	284	285	5.99	0.95
SimBench	94	20.2	9.79	15.5	0.77	0.95
case118†	118	13.7	98.3	98.8	7.22	47.47
case145†	145	493	3,538	3,650	7.40	1.35
iceland	189	39.8	13.7	15.3	0.38	1.89
illinois200†	200	10.5	14.9	14.9	1.42	5.95
case300†	300	110	124	128	1.16	2.47
LVN_heo1	722	52.8	3.39	8.5	0.16	0.88
case1354pegase†	1354	132	181	181	1.37	3.79
case1888rte	1888	142	52.5	53.7	0.38	1.87
GBnetwork	2224	112	41.5	46.2	0.41	0.65
case2848rte	2848	47.6	32	30.8	0.65	3.18
case2869pegase	2869	200	124	119	0.60	2.14
case3120sp	3120	236	61.9	57.5	0.24	1.35
ENTSO-E	6051	37.3	11.2	11.7	0.31	0.71
case6470rte	6470	130	23.6	25.6	0.20	1.60
case6495rte	6495	56.8	23	22.6	0.40	2.26
case6515rte	6515	88.6	24.7	26	0.29	1.80

Table S7. Floor rule across ranks (120 epochs). "Active": grids with $k < N$; "id.": grids with $k \geq N$ (CSP$\equiv C$). "Rule": active grids where raw>floor coincides with CSP lowering Mean PB. "CSP/floor": median of CSP_k Mean PB over the floor; values near 1 mean that CSP behaves like projecting the reference, so its effect is predictable from the floor. It is not a quality score.

k	Active	id.	PB$\downarrow$	$V\downarrow$	Both	Rule	CSP/floor
16	25	5	14	10	6	25/25	1.03
32	21	9	15	8	7	21/21	1.10
64	18	12	12	8	5	16/18	1.24
128	15	15	9	8	4	10/15	1.38